

Final Hardware Components List

Complete Working Cold-Chain Predictive Maintenance Prototype — ESP32-S3 + Peltier Primary/Backup Cooling

1. Sensing Components

Component	Specification	Qty.	Purpose	Approx. Cost
DS18B20 Waterproof Temperature Sensor	−55°C to +125°C, 1-Wire, waterproof probe	2	One for chamber temperature; one for Peltier hot-side/heatsink temperature	■160–240
SHT31 Temperature/Humidity Sensor	I²C temperature and relative-humidity sensor	1	Humidity monitoring and additional temperature measurement	■200–300
ACS712 Current Sensor	20A Hall-effect current sensor	1	Monitors primary Peltier current and helps detect abnormal operation	■100–150
Magnetic Reed Switch	NO/NC contact + magnet	1	Detects door opening and closing	■30–60
NEO-6M GPS Module	UART GPS receiver with antenna	1	Vehicle location, speed and timestamp for cloud monitoring	■250–400
Buzzer	5V active buzzer	1	Audible fault/alarm indication	■20–50
LEDs + Resistors	5mm LEDs with suitable resistors	2–3	Power, fault and backup-status indication	■30–50

2. Main Cooling System

Component	Specification	Qty.	Purpose	Approx. Cost
TEC1-12706 Peltier Module	12V, up to ~6A, 40×40mm	1	Primary cooling	■200–300
Aluminium Cold Plate	~40×40mm or slightly larger	1	Transfers cooling into the cold chamber	■200–350
Aluminium Heatsink	High-capacity, suitable for TEC1-12706	1	Removes heat from Peltier hot side	■300–500
12V DC Fan	40–60mm, suitable airflow	1	Cools the hot-side heatsink	■120–200
Thermal Paste	Standard thermal compound	1 tube	Improves thermal contact	■100–150
Thermal Pads	Suitable thickness	1 set	Improves thermal contact where required	■80–150

3. Backup Cooling System

Component	Specification	Qty.	Purpose	Approx. Cost
Small Backup Peltier	TEC1-12701 preferred; TEC1-12703 if unavailable	1	Backup cooling when primary cooling is declared failed	■120–220
Backup Mounting Hardware	Clamps/brackets as required	1 set	Secures backup Peltier in the cooling assembly	■100–150

4. Control and Processing Components

Component	Specification	Qty.	Purpose	Approx. Cost
ESP32-S3 Development Board	Wi-Fi/Bluetooth; flash/PSRAM preferred	1	Main ECU, sensor processing, local control and embedded ML inference	■1,000–1,400

Component	Specification	Qty.	Purpose	Approx. Cost
2-Channel Logic-Level MOSFET Driver	12V DC compatible; adequate current rating for Peltier loads	1	Independently switches primary and backup Peltier	■150–300
Perfboard / Prototype PCB	General-purpose PCB	1	Permanent interconnection and mounting	■50–100
Protection Components	Gate resistors, pull-downs and basic protection parts	1 set	Reliable MOSFET control and signal protection	■50–100
Power / Mode Switches	DC-rated	1–2	System power and manual control	■50–100

5. Power Supply Components

Component	Specification	Qty.	Purpose	Approx. Cost
12V SMPS	12V, 15A minimum	1	Powers Peltier(s) and fan	■900–1,200
Main Peltier Fuse + Holder	Approx. 7.5A; verify against selected Peltier	1	Protects primary Peltier circuit	■50–80
Backup Peltier Fuse + Holder	Approx. 2A; verify against selected Peltier	1	Protects backup circuit	■50–80
5V USB Power Adapter	Regulated 5V, ~1–2A	1	Separate power supply for ESP32	■150–250
Power Wires	16–18 AWG	As required	High-current Peltier wiring	■200–350
Signal Wires	26–28 AWG / jumper wires	As required	Sensors and control wiring	■100–180
Terminal Blocks / Connectors	Screw terminals, JST and DC connectors	1 set	Electrical connections	■150–250
Heat-shrink / Tape / Cable Ties	Assorted	1 set	Insulation and cable management	■150–250

6. Cold-Chain Box / Enclosure

Component	Specification	Qty.	Purpose	Approx. Cost
Mild Steel Sheet / Fabricated Box	~22 gauge; fabricated into prototype cube	1	Rigid outer structure	■500–900
Thermocol / EPS Insulation	40–50mm thickness	As required	Thermal insulation	■150–300
Foam/Rubber Door Gasket	Weather-strip type	1 set	Seals door and reduces heat leakage	■50–100
Metal Hinges	Small heavy-duty	2	Door movement	■50–100
Screws / Nuts / Washers	Suitable for steel sheet	1 set	Mechanical assembly	■150–250
Adhesive / Sealant	Suitable for EPS and steel	1 set	Fixes insulation and seals gaps	■100–150
Aluminium Brackets / Mounting Hardware	As required for cooling assembly	1 set	Mounts cold plate, Peltier and heatsink	■100–200

7. Cost Summary

Section	Estimated Low	Estimated High
Sensing components	■790	■1,250
Main cooling system	■1,000	■1,650
Backup cooling system	■220	■370
Control & processing	■1,300	■2,000
Power system & wiring	■1,750	■2,640
Cold-chain box / enclosure	■1,100	■2,000
Estimated complete prototype	■6,060	■9,910
Recommended practical budget	■7,500	■8,500
Safe maximum budget	—	■10,000

8. Components Deliberately Excluded

Refrigerant pressure transducer is excluded because this final prototype uses Peltier cooling and has no refrigerant loop. Also excluded from the base build are a second ACS712, second heatsink/fan/cold plate, vibration sensor, servo motor, 4-channel relay, nitrogen backup and LM2596 buck converter when a separate regulated 5V USB supply is used. GPS is retained because location-aware cloud monitoring is part of the complete project concept.

9. Final Prototype Operation

Normal: Sensors → ESP32-S3 → local processing/ML → primary Peltier ON.

Fault: abnormal temperature/current/hot-side behaviour → primary Peltier OFF → backup Peltier ON → alarm/status indication.

Cloud: ESP32-S3 sends sensor, prediction, fault and GPS information to the cloud dashboard.

10. Important Purchasing Note

The cost figures are planning estimates rather than fixed quotations. Actual prices vary by supplier, component quality, shipping and local fabrication. Before purchase, verify that the selected MOSFET driver can continuously handle the actual Peltier current and that the 12V/15A SMPS has adequate continuous capacity.